
(a) State in Continuous $\mathbb{R}^2$
Work Remaining $W_{j,i}$ (norm)
Earliest Start Time $ST_{j,i}$ (norm)
$O_{1,1}$
$O_{1,2}$
$O_{2,1}$
$O_{2,2}$
$O_{3,1}$
$O_{3,2}$
Operations mapped to (ST, W) coordinates
Node size ∝ Duration
(b) 2D Priority Field Canvas
Bilinear Splatting
Work Dimension (Y)
Start Time Dimension (X)
Continuous Density Field
Continuous Field [M, H, W]
(c) U-Net Spatial Reasoning
ConvBlock 64x64
/2 Pool
ConvBlock 32x32
Skip Conn
UpBlock 32x32
UpBlock 64x64 (Field)
Bottleneck + Machine Attention
Receptive field scales with O(M · HW) vs O(N²) Graph
(d) Bilinear Readout → Action
Bilinear Querying
Policy Probability π(a | s)
Selected Action (Bilinear Readout)
68.0%
14.0%
12.0%
6.0%
$(O_{1,1}, M_1)$
$(O_{1,1}, M_2)$
$(O_{2,1}, M_1)$
$(O_{3,1}, M_3)$